8.9.1 Determination of solubility product from molar solubility

Solubility product can be calculated from the molar solubility i.e., the maximum number of moles of solute that can be dissolved in one litre of the solution.

For a solute X_mY_n,

$$X_mY_n(s) \rightleftharpoons mX^{n+}(aq) + nY^{m-}(aq)$$

From the above stoichiometrically balanced equation we have come to know that 1 mole of $X_mY_n(s)$ dissociated to furnish 'm' moles of X^{n+} and 'n' moles of Y^{m-} if 's' is molar solubility of X_mY_n, then

$[X^{n+}]=ms$ and $[Y^{m-}]=ns$

$\therefore K_{sp} = [X^{n+}]^m [Y^{m-}]^n$

$K_{sp}=(ms)^m(ns)^n$

$K_{sp}=(m)^m(n)^n(s)^{m+n}$

Example 8.10

- Establish a relationship between the solubility product and molar solubility for the following

a) $BaSO_4$ b) $Ag_2(CrO_4)$

$BaSO_4(s) \overset{H_2O}{\rightleftharpoons} Ba^{2+}(aq)+SO_4^{2-}(aq)$

$K_{sp}=[Ba^{2+}][SO_4^{2-}]$

$= (s)\ (s)$

$K_{sp} = S^2$

$Ag_2CrO_4(s) \overset{H_2O}{\rightleftharpoons} 2Ag^+(aq)+CrO_4^{2-}(aq)$

$K_{sp}=[Ag^+]^2[CrO_4^{2-}]$

$= (2s)^2(s)$

$K_{sp}=4s^3$

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